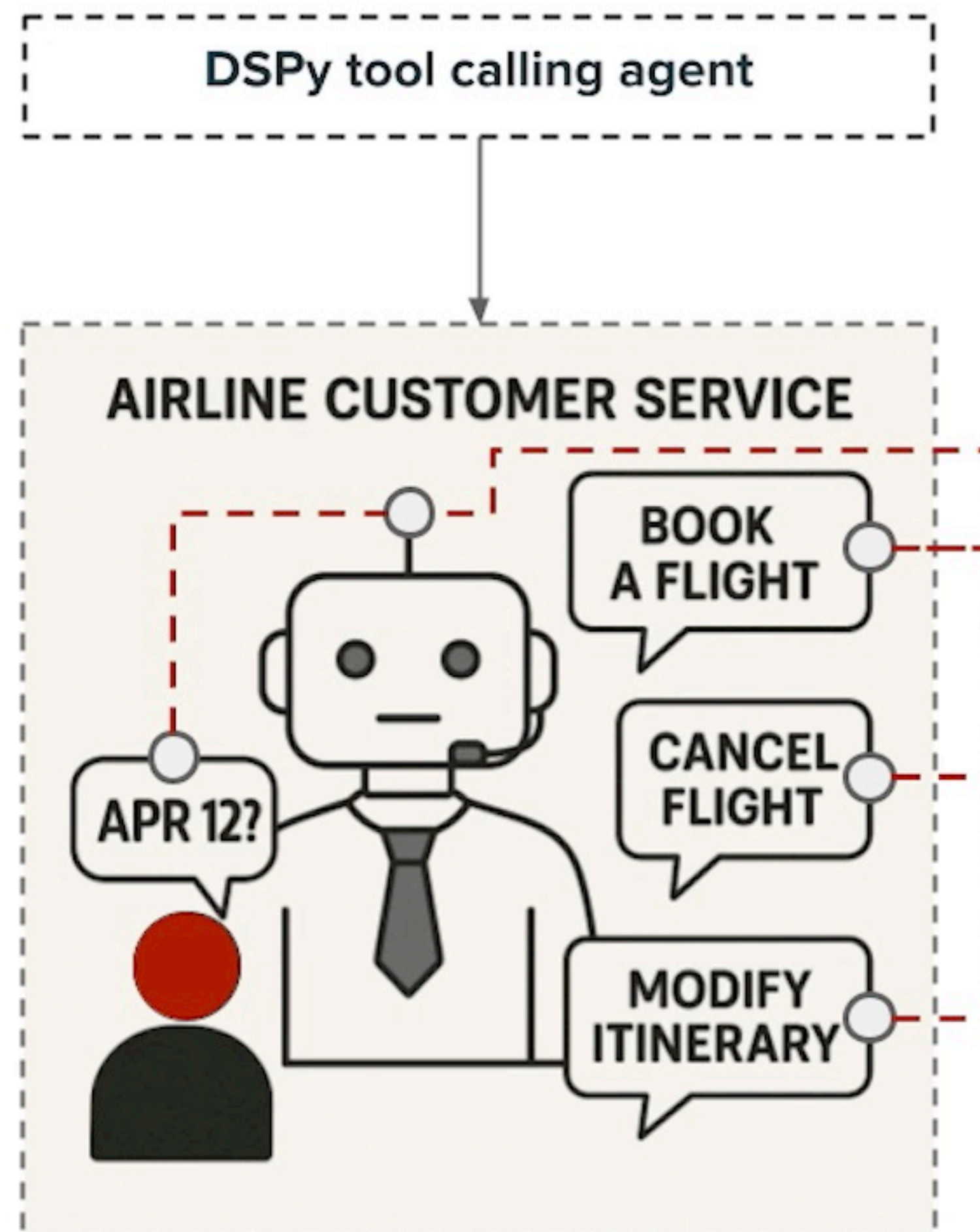


Use Managed MLflow Service

Databricks lighthouse: signup.databricks.com

Lab Materials



What is tracing, and why tracing?

- **Tracing:** Records **inputs and outputs** of intermediate steps and captures **hierarchical call stack**
- **GenAI apps are complex:** Internal logic is often opaque
- **Without tracing: One failed LM** call among many can break the whole program

Tracing improves:

- ✓ Interpretability
- ✓ Debugging

What is MLflow

Open source MLOps framework

- Supports **ML and GenAI app development**
- Covers the **full lifecycle**: development, tracking, deployment

Focus on reproducibility and traceability

- Manage every experiment, run, and model version with ease

Open source server + client

- Fully open and extensible

Get Started with Tracing DSPy

```
import mlflow

mlflow.dspy.autolog() → This is all you need!

class MyProgram(dspy.Module):
    def __init__(self, **kwargs):
        super().__init__()

        self.module1 = dspy.ChainOfThought(MySignature1)
        self.module2 = dspy.ChainOfThought(MySignature2)

    def forward(self, inputs, **kwargs):
        x = preprocess(x)
        x = self.module1(inputs)
        x = midprocess(x)
        x = self.module2(x)
        return postprocess(x)

my_program = MyProgram()

my_program(inputs="what's the meaning of life")
```

MLflow Trace UI

The screenshot displays the MLflow 2.21.3 interface. The top navigation bar includes 'Experiments', 'Models', and 'Prompts'. The left sidebar shows the 'Experiments' section with a search bar and a list of experiments: 'Default' and 'dspy_workshop'. The 'dspy_workshop' experiment is selected. The main content area shows the 'dspy_workshop' experiment details, including a 'Provide Feedback' link and an 'Add Description' button. Below this, there are tabs for 'Runs', 'Evaluation', 'Experimental', and 'Traces'. The 'Traces' tab is active, showing a table of traces. The table has columns for 'Request ID', 'Time created', 'Status', 'Request', 'Response', and 'Run name'. A single trace is listed with a status of 'OK'.

Request ID	Time created	Status	Request	Response	Run name
858dea58bb96480eb82a0...	5 days ago	OK	{ "user_request": "help me book a flight from SFO to JFK on 09/01/2025, my name is Adam" }	{ "trajectory": { "thought": "I need to fetch flight information for Adam from SFO to"	

Traces are linked to MLflow experiments.

MLflow DSPy Trace Structure

The following DSPy components are traced:

- Module
- Adapter
- LM
- Tool

ReAct.forward

×

Task name	Inputs / Outputs	Attributes	Events
ReAct.forward 13.48s			
Predict.forward_1 2.20s			
ChatAdapter.format_1 0.01s			
LM.__call__1 2.19s			
LM.__call__2 2.19s			
ChatAdapter.parse_1 0.01s			
Tool.fetch_flight_info 0.01s			
Predict.forward_2 2.85s			
ChatAdapter.format_2 0.01s			
LM.__call__3 2.85s			
LM.__call__4 2.85s			
ChatAdapter.parse_2 0.01s			
Tool.pick_flight_1 0.01s			
.forward_3 2.58s			

Inputs

user_request
please help me book a flight from SFO to JFK on 09/01/2025, my name is Adam

Outputs

trajectory

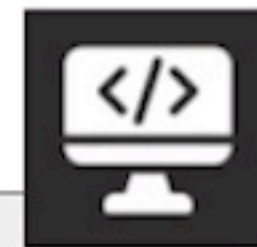
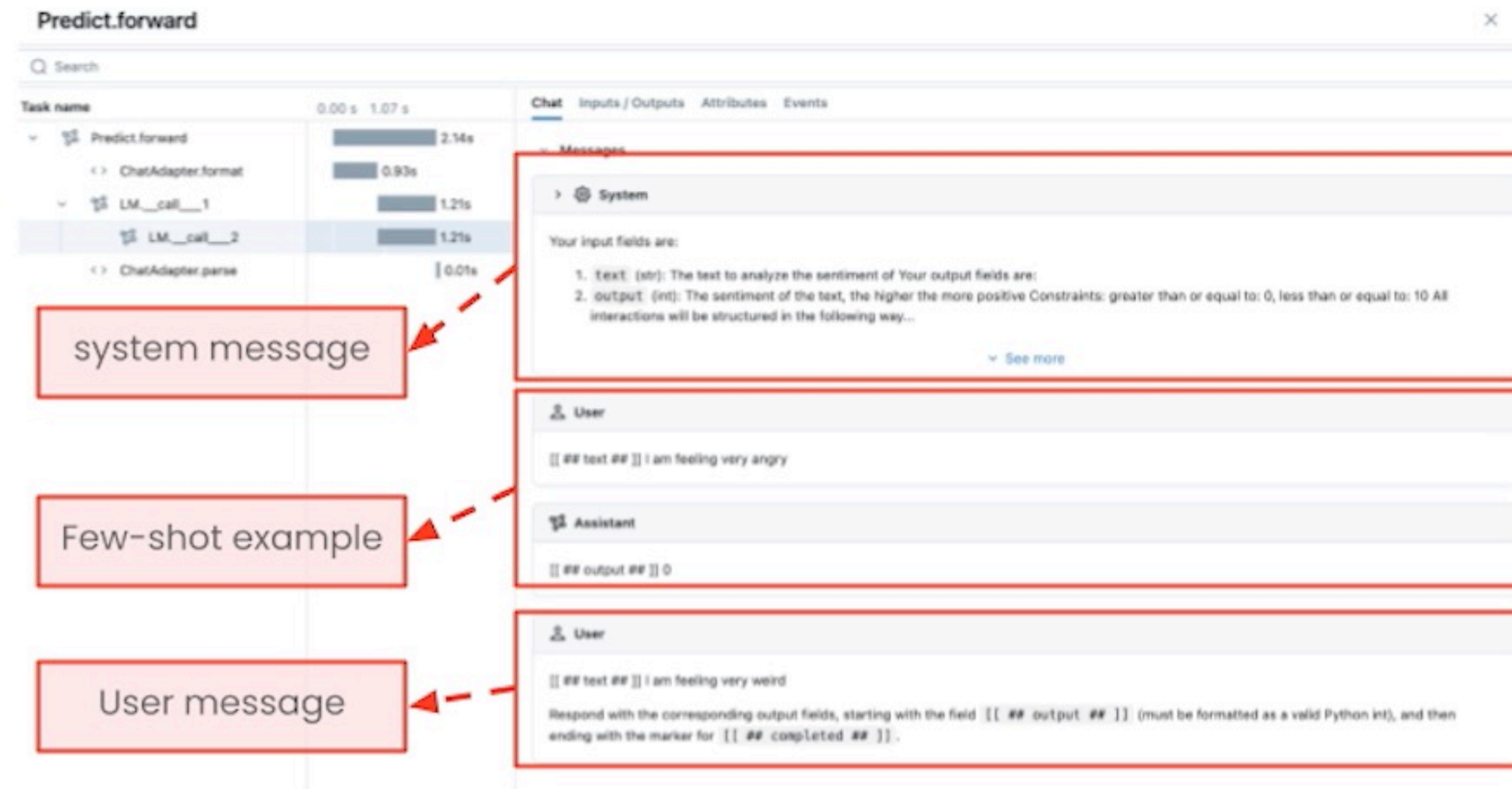
```
1 {
2   "thought_0": "I need to fetch flight information for Adam from SFO to JFK
3   "tool_name_0": "fetch_flight_info",
4   "tool_args_0": {

```

See more

reasoning

Use MLflow Trace to Interpret Prompt



Let's see all of this in the lab